RAYNAUD-GRUSON FLATTENING OF A FINITE TYPE MODULE ON A SPACE BY A U-ADMISSIBLE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8777-8794 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8796-8846, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author's first paragraph of the schemes-side argument (More on Flatness) is quoted inside the chapter proof as the driver; no bibliography reference is invoked.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale descent of strict transforms and of flatness/finite presentation (Divisors on Spaces, strict-transform-local, strict-transform-flat; Properties of Spaces descent), the schemes Raynaud-Gruson flattening theorem (More on Flatness, theorem-flatten-module), and the ideal-pushdown lemma (lemma-push-ideal, same chapter).

Source reference. Original theorem 0815: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8846–8846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 087B: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8698–8712. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 085Y: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8846–8846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 085S: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8846–8846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0868: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8846–8846. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0865: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8846–8846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0863: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8846–8846. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *Let S be a scheme. Let B be a quasi-compact and quasi-separated algebraic space over S. Let X be an algebraic space over B. Let $\mathcal{F}$ be a quasi-coherent module on X. Let $U \subset B$ be a quasi-compact open subspace. Assume*

(1) *X is quasi-compact,*
(2) *X is locally of finite presentation over B,*
(3) *$\mathcal{F}$ is a module of finite type,*
(4) *$\mathcal{F}_U$ is of finite presentation, and*
(5) *$\mathcal{F}_U$ is flat over U.*

Then there exists a U-admissible blowup $B' \to B$ such that the strict transform $\mathcal{F}'$ of $\mathcal{F}$ is an $\mathcal{O}_{X \times_B B'}$-module of finite presentation and flat over B'.

Proof. Choose an affine scheme V and a surjective étale morphism $V \to X$. Because strict transform commutes with étale localization (Divisors on Spaces, Lemma 0863) it suffices to prove the result with X replaced by V. Hence we may assume that $X \to B$ is representable (in addition to the hypotheses of the lemma).

Assume that $X \to B$ is representable. Choose an affine scheme W and a surjective étale morphism $\varphi : W \to B$. Note that $X \times_B W$ is a scheme. By the case of schemes (More on Flatness, Theorem 0815) we can find a finite type quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_W$ such that (a) $|V(\mathcal{I})| \cap |\varphi^{-1}(U)| = \emptyset$ and (b) the strict transform of $\mathcal{F}|_{X \times_B W}$ with respect to the blowing up $W' \to W$ in $\mathcal{I}$ becomes flat over W' and is a module of finite presentation. Choose a finite type sheaf of ideals $\mathcal{J} \subset \mathcal{O}_B$ as in Lemma 087B. Let $B' \to B$ be the blowing up of $\mathcal{J}$. We claim that this blowup works. Namely, it is clear that $B' \to B$ is U-admissible by our choice of ideal $\mathcal{J}$. Moreover, the base change $B' \times_B W \to W$ is the blowup of W in $\varphi^{-1}\mathcal{J} = \mathcal{I}\mathcal{I}'$ (compatibility of blowup with flat base change, see Divisors on Spaces, Lemma 085S). Hence there is a factorization

$$W \times_B B' \to W' \to W$$

where the first morphism is a blowup as well, see Divisors on Spaces, Lemma 085Y). The restriction of $\mathcal{F}'$ (which lives on $B' \times_B X$) to $W \times_B B' \times_B X$ is the strict transform of $\mathcal{F}|_{X \times_B W}$ (Divisors on Spaces, Lemma 0863) and hence is the twice repeated strict transform of $\mathcal{F}|_{X \times_B W}$ by the two blowups displayed above (Divisors on Spaces, Lemma 0868). After the first blowup our sheaf is already flat over the base and of finite presentation (by construction). Whence this holds after the second strict transform as well (since this is a pullback by Divisors on Spaces, Lemma 0865). Thus we see that the restriction of $\mathcal{F}'$ to an étale cover of $B' \times_B X$ has the desired properties and the theorem is proved. □